To D or Not to D: A Comparative Analysis of Visual SLAM

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Introduction

Modern SLAM systems are capable of achieving accurate robot localization by leveraging depth sensors. But the inclusion of more sensors means an increase in hardware cost, power, and complexity. On the other hand, monocular SLAM systems use only a single camera but suffer from problems such as scale ambiguity and drift. This raises the question "Can learned depth priors replace depth sensors in modern SLAM systems while maintaining the same level of accuracy?". Recent advances in research, such as the paper "Dropping the D: RGB-D SLAM Without the Depth Sensor" [1], have tackled this problem. However, a comparison between this method and more classical methods has not been done yet. Therefore, we have implemented a "lite" version of the DropD system that replaces physical depth sensors with a pretrained monocular depth model and integrated it into an RGB SLAM system. We then compared this approach against more classical SLAM systems, such as FAST-LIO2 [2] and ORB-SLAM3 [3], to perform an analysis of how the learned depth priors stack up against these older methods.

Related Work

- **LiDAR-Inertial SLAM:** LiDAR-Inertial Odometry (LIO), such as FAST-LIO2 [2], directly register raw point clouds using highly efficient structures and utilize Kalman filters for robust, scale-accurate estimation. The active ranging provides metric depth natively, eliminating scale ambiguity entirely. However, this requires expensive sensing hardware.
- **Visual SLAM:** Feature-based visual pipelines, such as ORB-SLAM [3], set the standard for geometric accuracy. However, they are sensitive to dynamic objects and monocular systems suffer from scale ambiguity since depth cannot be recovered from a single image without additional priors, yielding trajectories that are correct up to an unknown scale factor.
- **Dynamic Environments:** Moving objects severely disrupt reliable post estimation. With the rise of lightweight segmentation models, such as YOLOv11 [4], dynamic objects can be masked in real-time, preventing moving features from corrupting the pose graph.
- **Learned Priors in Monocular SLAM:** Modern deep learning models can be used to replace traditional SLAM components. For example, UniDepthV2 [5] produces dense metric depth for scale recovery from a single image and Key.Net [6] improves feature repeatability under motion blur.

References

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Methodology

Experimental Setup

- **Hardware:** Laptop w/ AMD Ryzen AI 7 CPU, NVIDIA RTX 5070, 32GB RAM.
- **Benchmarks: TUM RGB-D** (*fr1/desk2, fr2/desk, fr3/structure_texture_near*) [7] and **KITTI Odometry** (Scenes *00-10*) [8]

ORB-SLAM3 [3]

- Evaluated as the baseline for purely geometric SLAM, utilizing a multi-threaded architecture (Tracking, Local Mapping, and Loop Closing/Map Merging).
- Uses FAST corner detection and ORB descriptors.
- **Limitations:** Assumes static world; lacks monocular metric scale.

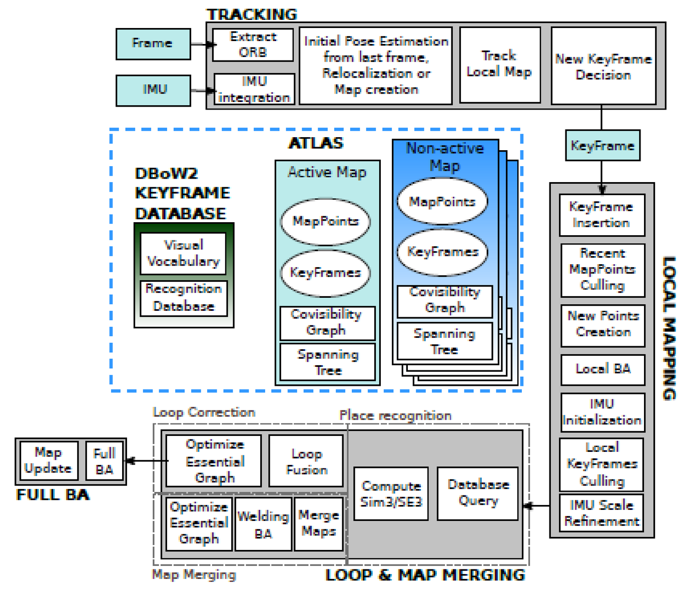

Figure 1. ORB-SLAM3 system [3].

DropD-SLAM [1]

Processes images through 3 NN then into unmodified ORB-SLAM3 back-end:

1. **Learned Feature Extraction:** Key.Net [6] extracts repeatable 2D keypoints paired with ORB descriptors.
2. **Dynamic Masking:** A YOLOv11 [4] creates instance masks that are dilated so dynamic keypoints can be discarded.
3. **Metric Depth Association:** UniDepthV2 [5] assigns absolute scale to the image.

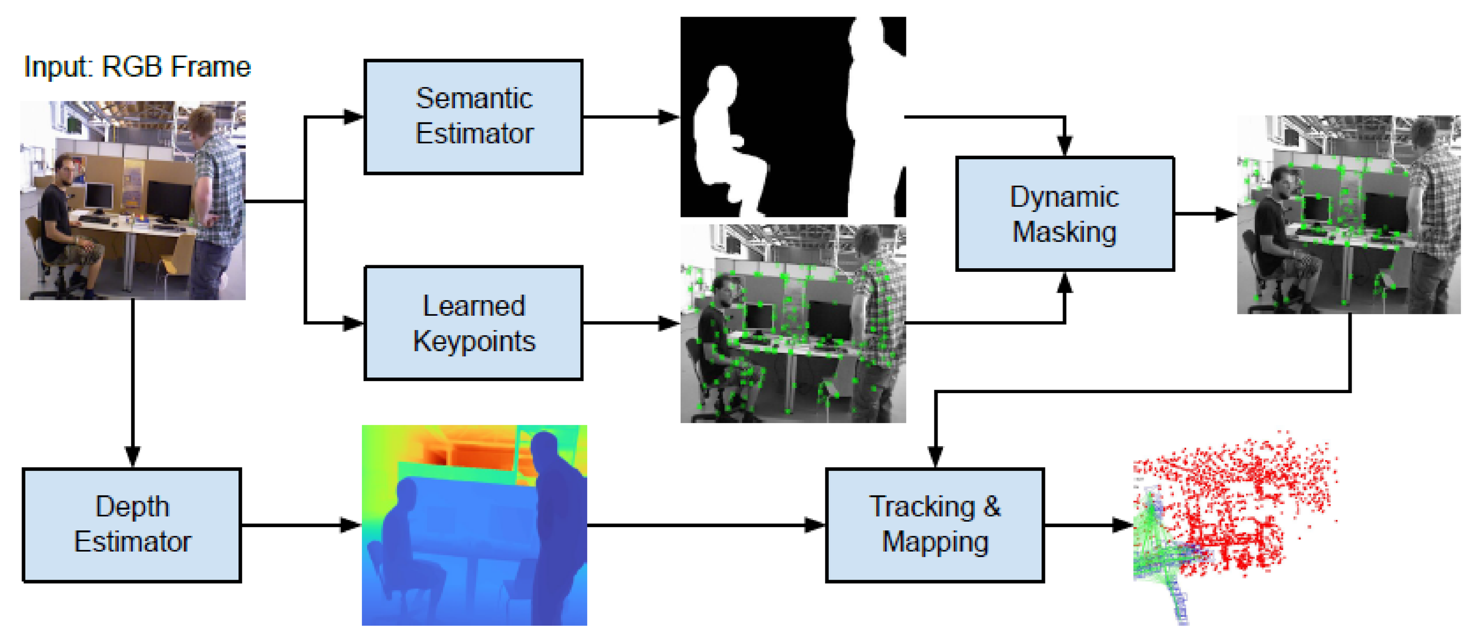

Figure 2. Overview of the DropD-SLAM system architecture. Adapted from [1].

FAST-LIO2

- Evaluated as LiDAR-Inertial baseline. Bypasses feature extraction entirely.
- Tightly couples high-frequency IMU data with raw point clouds.
- Utilizes an ikd-Tree to efficiently register raw points to the global map.
- **Advantages:** Delivers highly robust, scale-accurate state estimation regardless of dynamic objects or lighting conditions.

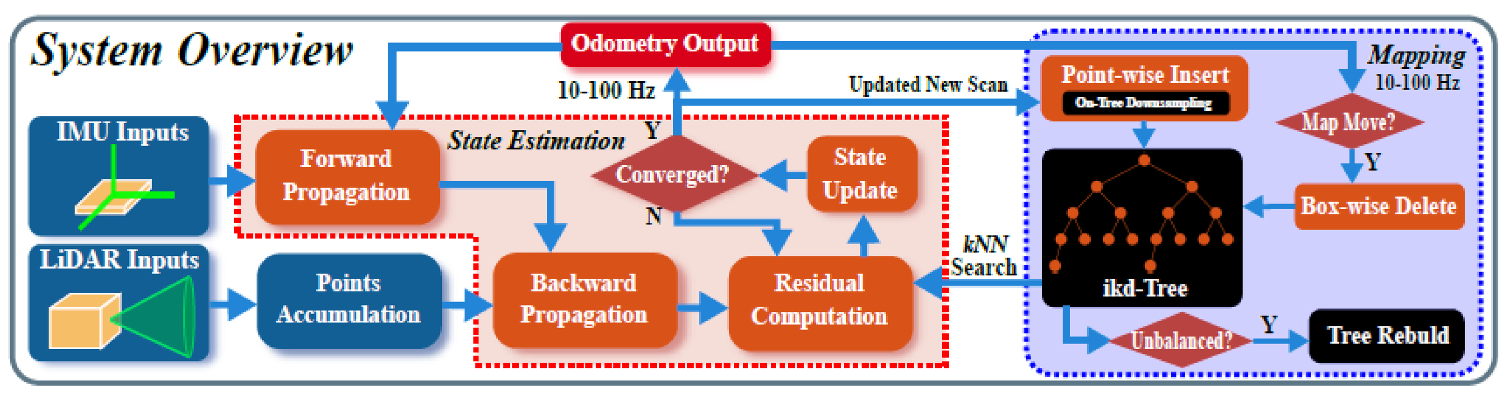

Figure 3. Overview of the FAST-LIO2 system architecture. Adapted from [2].

Results

Our comparison was based on several metrics:

- **Total Frames Tracked:** tells us if the system was able to keep track of the ego vehicle throughout the sequence
- **Absolute Pose Error:** to measure the global drift of the full trajectory, raw and aligned
- **Relative Pose Error:** to measure local accuracy or how much error accumulates over smaller frame segments of the sequence
- **Drift Rate:** shows us the normalized summary of how fast the trajectory diverges
- **Map Quality:** gives us how dense the built map was at the end of the sequence
- **Runtime Performance:** indicates if the system can run in real-time

Overall, DropD-SLAM and its variants had very high tracking success, only missing a few frames. ORB-SLAM3 never missed a frame in any scene. But FAST-LIO2 had more trouble, only able to successfully track 70% of the frames.
We share our most important findings here. DropD-SLAM suffers a lot from scale drift but if that can be corrected for, it is very accurate. From Figure 4, DropD-SLAM has low error that comes close to ORB-SLAM3. Furthermore on the TUM datasets, DropD-SLAM's absolute pose error got much closer to ORB-SLAM3's, suggesting that ORB-SLAM3's advantage on KITTI comes primarily from loop closure on long outdoor sequences. On a short, loopless indoor path, the two systems are likely comparable in accuracy.

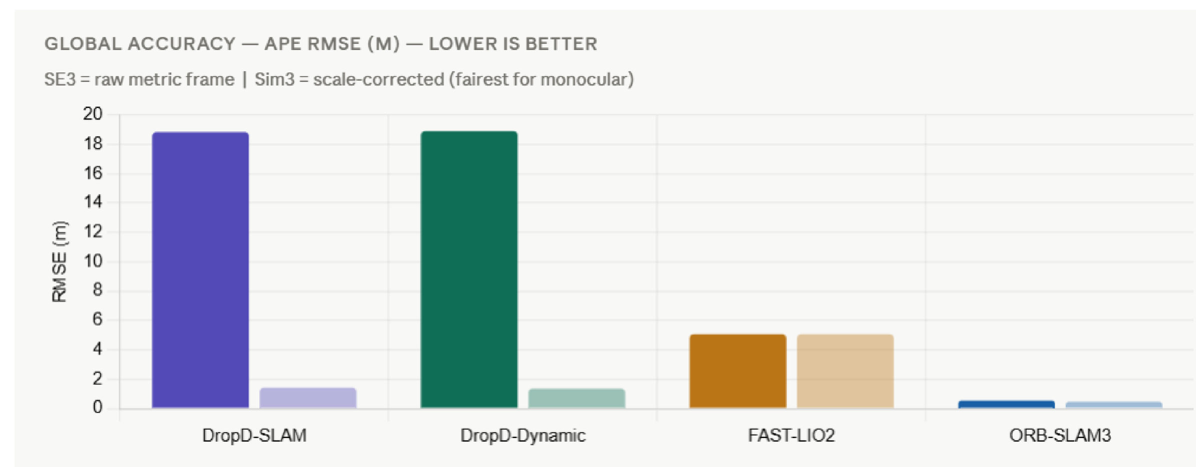

Figure 4. Average absolute pose error for all methods.

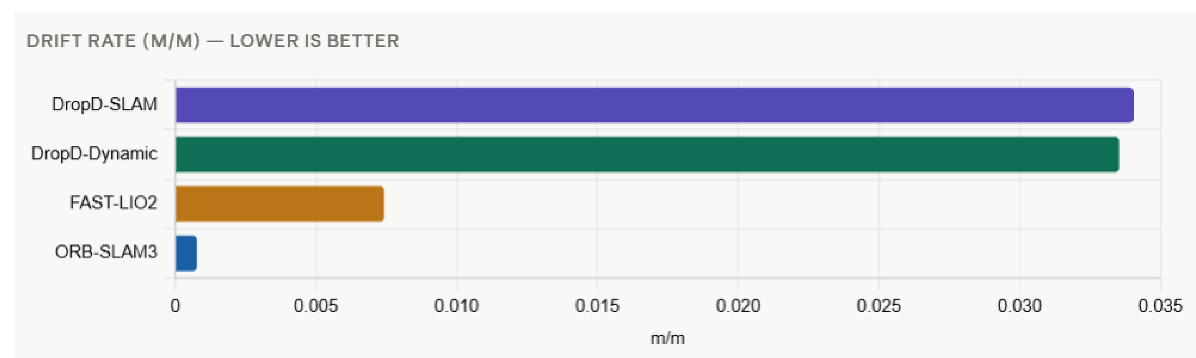

Figure 5. Average drift rate for all methods.

Conclusion

Using a pretrained monocular depth model instead of physical depth sensors seems to be quite effective. It is locally accurate, barely loses track, and produces geometrically reasonable trajectories. However, it has a large weakness in global scale correction. On short sequences or scenes without loop closure opportunities, it is competitive with ORB-SLAM3. But on long sequences where accumulated scale drift matters, it falls behind ORB-SLAM3 noticeably.